Experiment 2: Automating ML Training/Testing using GitHub Actions with Azure ML Integration

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AIM

To study the automation of Machine Learning training and testing using GitHub Actions and its integration with Azure ML pipelines.

OBJECTIVES

1. To understand CI/CD concepts in Machine Learning projects.
2. To study offline setup using GitHub Desktop and YAML-based CI configuration.
3. To understand integration of GitHub Actions with Azure ML pipeline for automated cloud execution.

THEORY

1. CI/CD in Machine Learning

Continuous Integration (CI) and Continuous Deployment (CD) automate building, testing, and deployment processes. In ML projects, automation ensures that whenever code is updated, model training and testing are triggered automatically.

This reduces manual effort, improves consistency, and speeds up development.

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2. Offline Setup: GitHub Desktop and YAML

GitHub provides tools to manage repositories and workflows.

GitHub Desktop is used to manage code locally and push updates. CI workflows are defined using YAML files, which specify automated steps such as installing dependencies, training, and testing models.

This ensures a structured and consistent local workflow.

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3. Online Setup: GitHub Actions

GitHub Actions automatically runs workflows when code is pushed or updated.

It executes tasks in virtual environments, enabling automated model validation and testing. This improves collaboration and ensures code quality before deployment.

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4. Integration with Azure ML Pipeline

Microsoft Azure provides

Azure Machine Learning for scalable ML workflows.

GitHub Actions can trigger Azure ML pipelines for automated data processing, training, and evaluation in the cloud.

This integration enables scalable, consistent, and fully automated ML lifecycle management.

CONCLUSION

In this experiment, we studied how GitHub Actions automates ML training and testing and integrates with Azure ML pipelines.

This approach improves efficiency, reproducibility, and scalability in modern ML workflows.